

FIFTH SEMESTER DIPLOMA EXAMINATION IN QS & CM**MODEL QUESTION PAPER****ESTIMATING AND COSTING****Time -3hrs****(MAXIMUM MARKS -100)**

[Note: 1.The missing data may be suitably assumed

2. Quantities to be worked out in tabulated form

3. sketch to be accompanied]

PART A

I. Answer the following in one or two sentences 2 marks each.

1. Find out the weight per meter of a reinforcing bar having diameter 16 mm.
2. State the purpose of specification .
3. What do you mean by soling coat in road construction ?
4. Define salvage value .
5. Write any four method of valuation .

5x2=10

PART B

II. Answer any five questions from the following 6 marks each

1. For the given building find out the quantity of RCC for lintel . The size of lintel is 20cm x 15cm and it is provided throughout the wall .
2. In a RCC beam of span 7m and 400 x750 mm size having the following steel reinforcement

Main bar : 22 mm dia straight bar @ 2.98 kg /m 3 nos

20 mm dia bent up bar @ 2.47 kg/m 3nos

Stirrup holder : 8 mm dia @ .62 kg/m 18 nos

6 mm dia @ .22 kg/m 14 nos

Prepare bar bending schedule .

3. Write down the detailed specification of pointing with cement mortar 1 : 3
4. Write any 6 essential requirements of a wood specification
5. A portion of road having 150 m length , in an uniform ground. The height of bank at the two ends being 1.2 m and 1.7 m . the formation width is 10 m and side slope 2 : 1 . Assume there is no transverse slope .find out the quantity of earth work
 1. Using mean sectional area method .
 2. Prismoidal method .

6. Write any six purpose of valuation .
7. Describe the different methods of valuation

6x5=30

PART C

Answer any one full question from each module

MODULE I

- III. For the column footing shown in the attached figure , find out the quantities of :
1. Earth work excavation
 2. Plain cement concrete 1 : 4: 8
 3. Reinforced cement concrete 1 : 1 ½ : 3
 4. Steel reinforcement

15

OR

- IV. From the accompanying figure find out the quantities of RCC for
1. Roof slab
 2. Lintel & sun shade

MODULE II

- V.a. Write the detailed specification for RCC 1 : 1 ½ : 3 using 20 mm nominal size broken stone for RCC Beam 8
- b. Write detailed specification of plastering on walls using cement mortar 1: 4 , 12 mm thick one coat 7

OR

- VI. a. Differentiate between general specification and detailed specification 5
- b. For a brick masonry wall in cement mortar 1: 6 , write down General & Detailed specification 10

MODULE III

- VII. a. Draw the tabular to calculate the quantity of Earth work using Prismoidal formula 4
- b. How you will calculate the quantity of earth work using Prismoidal formula when there is an even number of areas 3
- c. A road embankment 6 m wide at formation level with side slope 2:1 is formed 80 m contour to 224 m contour with an average gradient of 1 in 50 and height of bank of 2.4 m .
Compute the volume of the embankment . 8

OR

- VIII. Compute the quantity of earth work involved in the formation of a road 10m wide with side slope 1 ½ : 1 for a length of 500 m from the data given .Ground may be assumed level in the transverse direction
- | | | | | | | | |
|----------|---|---|-----|-----|-----|-----|-----|
| Distance | : | 0 | 100 | 200 | 300 | 400 | 500 |
|----------|---|---|-----|-----|-----|-----|-----|

Ground level : 100.8 100.3 102.1 102.5 103.2 104.2

P.T.O

Formation level : 101.00 up gradient 1 in 200 upto 300 m and then horizontal upto 500

15

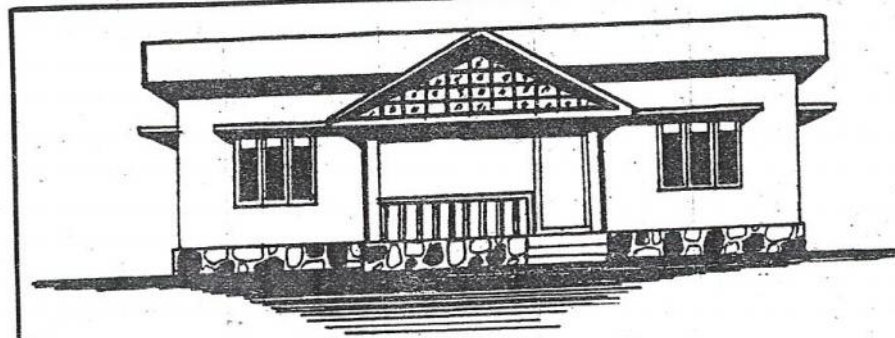
MODULE IV

- IX.** a. Explain any one method of calculating depreciation of a property 5
b. A residential building constructed 8 years ago is situated on a plot whose total area is 420 sqmeters . The plinth area of the building is 220 sqmeters .The present cost of construction is 12 lakhs . The cost of land is 20,000 per sqmeter . the rate of depreciation for the value of the building is 1% .Calculate the total value of the property 10

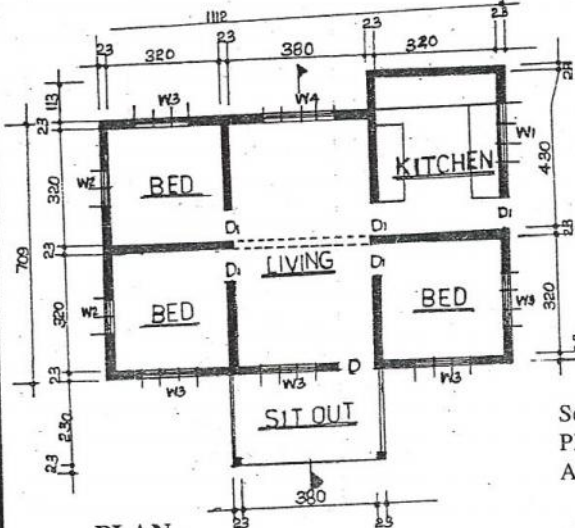
OR

- X.** a. Write short note on various methods of rent fixation 7
b. the gross rent of a property is Rs 5000 per month . The outgoings are 35% of gross rent .The highest rate of interest is 5% .Determine the capital cost value of the property 8

4010 (1)



ELEVATION

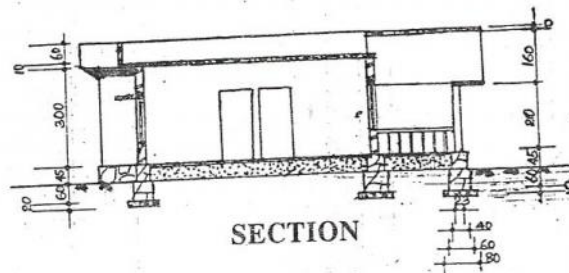


PLAN

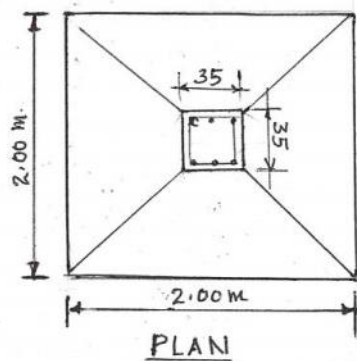
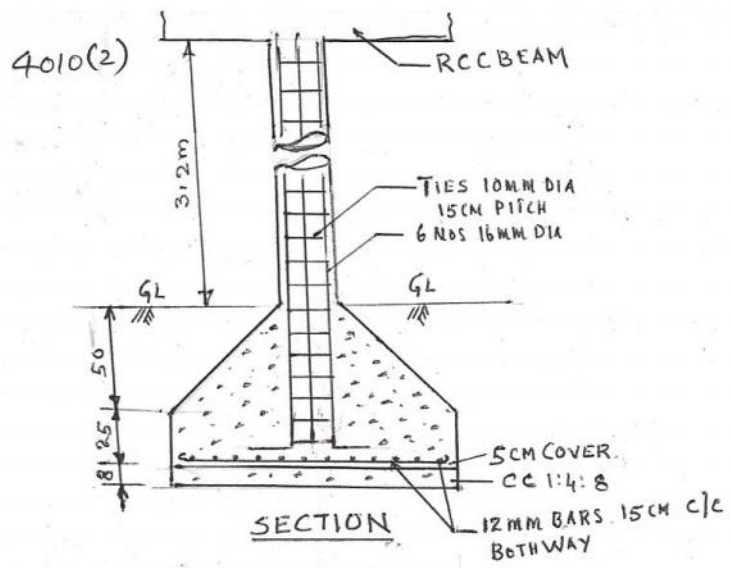
Specification

Door	D	100 x 210
"	D ₁	90 x 210
Window	W ₁	150 x 200
"	W ₂	100 x 135
"	W ₃	150 X135
"	W ₄	200 X135

Scale: 1:100
Plinth Area 93.64 m²
All Dimensions are in Cm



SECTION



RCC COLUMN FOOTING